



CD8⁺ Memory T Cells Encode Obesogenic Memory in White Adipose Tissue

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Obesity is associated with systemic energy imbalance and leads to a wide range of metabolic complications, including type 2 diabetes, fatty liver disease, and cardiovascular disorders. Therapeutic strategies for obesity, including lifestyle interventions such as exercise and dietary modification, bariatric surgery, and glucagon-like peptide 1 receptor agonists, have been widely implemented (1). However, rapid weight regain is frequently observed after cessation of weight management medications (2). Moreover, although weight loss (WL) induced by these strategies improves many metabolic phenotypes, subsequent weight regain, referred to as weight cycling (WC), has been reported to exacerbate metabolic dysfunction even more than stable obesity (1). Given that WC commonly occurs in daily life, substantial efforts have been made to understand its underlying mechanisms and to develop strategies to prevent its detrimental consequences.

The exacerbation of metabolic phenotypes during WC suggests that metabolic or obesogenic “memory” might be stored within the organism, recalled upon weight regain, and thereby contribute to metabolic deterioration (3). Adipose tissue is a central organ in systemic energy metabolism and is among the first tissues to respond to obesogenic stimuli, undergoing extensive remodeling in size, cellular composition, and functional states (4,5). These features make adipose tissue an ideal candidate site for metabolic memory. Such memory of prior obesity could be encoded as altered cellular composition (3), epigenetic reprogramming within specific cell types (6), and immune cell memory (e.g., T-cell memory) (7).

Adipose tissue is composed of various cell types, including adipocytes as the parenchymal cell type, adipocyte progenitors, endothelial cells, and diverse immune cell populations. During obesity, proinflammatory cell types such as lipid-associated macrophages, Th1 CD4⁺ T cells, and CD8⁺ T cells accumulate in adipose tissue, exacerbating

chronic inflammation and reshaping tissue microenvironment (4,8). Previous studies have shown that obesity-induced alterations, including epigenetic reprogramming of adipocytes (6) and retention of macrophages (9), can persist during WL and contribute to accelerated metabolic deterioration upon weight regain. However, the identities of the cellular mediators of metabolic memory in WC, and the underlying mechanisms by which such memory is established, are largely unknown.

In this issue of *Diabetes*, Garcia et al. (10) propose that CD8⁺ T cells act as key contributors to metabolic memory under WC conditions. Using single-cell RNA sequencing and flow cytometry across lean, obese, WL, and WC states, the authors show that CD8⁺ effector memory T (T_{EM}) cells are enriched in adipose tissue during obesity, WL, and especially WC. Single-cell T-cell receptor sequencing further revealed the accumulation of highly clonal CD8⁺ T_{EM} cell populations in the WC group. Importantly, genetic deletion of CD70, which disrupts the CD27–CD70 costimulatory axis, impaired CD8⁺ T_{EM} cell formation and abolished clonal expansion, thereby alleviating glucose intolerance observed during WC. Mechanistically, ligand–receptor interactions between T cells, macrophages, and monocytes were diminished due to CD70 deletion under WC conditions. Together, these findings suggest that CD8⁺ T_{EM} cells play a critical role in mediating metabolic memory (Fig. 1).

CD4⁺ and CD8⁺ T cells differ substantially in their memory formation pathways and functional heterogeneity. For example, CD4⁺ T-cell memory is less dependent on CD70–CD27 signaling and is divided into diverse subsets, including Th1, Th2, and Th17 populations, whereas CD8⁺ T-cell memory is strongly impaired by CD70 deletion (11). Garcia et al. demonstrate that CD70 deficiency selectively disrupts CD8⁺ T cell memory formation, while Th1 CD4⁺ memory T cells are unchanged or relatively increased (10). These observations suggest that CD4⁺

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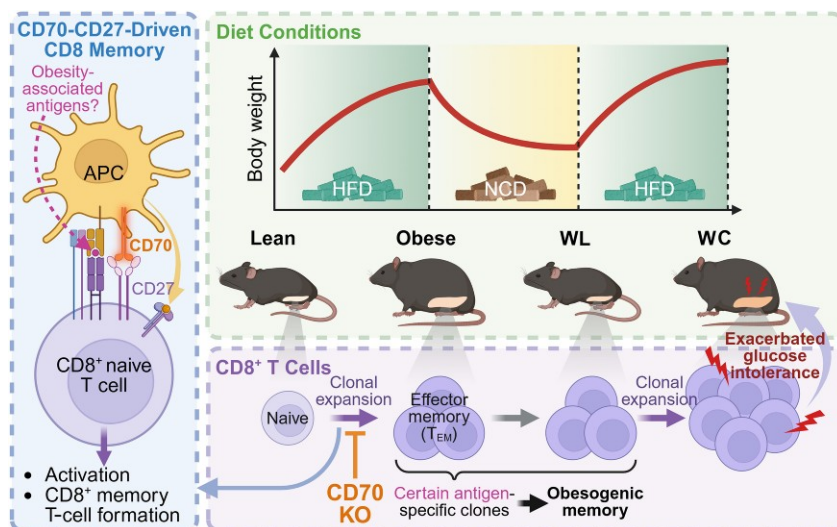


Figure 1—Proposed model of CD8⁺ memory T-cell accumulation in epididymal WAT during iterative obesogenic stress. Obesity-associated antigens induce clonal expansion of specific CD8⁺ memory T cells. During subsequent WL and obesogenic challenges, these cells expand further and contribute to worsening glucose homeostasis under WC conditions. These processes are markedly attenuated in CD70-deficient mice because of impaired CD8⁺ memory T-cell generation and expansion in obesity. HFD, high-fat diet; KO, knockout; NCD, normal chow diet.

T cells may not be primary drivers of obesogenic memory and might even exert protective roles. However, given prior reports suggesting CD4⁺ T cells as a memory cell type during weight regain (12) and the possibility of CD70-dependent autonomous effects, these interpretations should be made with caution. As the authors noted, this could be further clarified by using an antigen-presenting cell (APC)-specific CD70 deletion system.

In obese adipose tissue, CD8⁺ T cells interact with various cell types, including macrophages, adipocytes, and adipocyte progenitors, via antigen presentation and cytokines such as CCL5, IFN- γ , and tumor necrosis factor- α (13–16). Depletion of CD8⁺ T cells mitigates obesity-induced macrophage infiltration, polarization, and chronic inflammation in adipose tissue (15). Although Garcia et al. show potential interactions between T cells, macrophages, and monocytes under WC conditions, the precise cellular and molecular mechanisms by which accumulated CD8⁺ T_{EM} cells drive metabolic deterioration in WC remain to be elucidated. Histological analyses and coculture experiments would further clarify whether and how these cellular interactions occur in adipose tissue during WC.

This study provides valuable single-cell RNA sequencing data sets that map how adipose tissue immune cell populations are shaped across lean, obese, WL, and WC conditions in both wild-type and CD70-deficient mice. When combined with functional assays such as the glucose tolerance test and flow cytometry, the authors convincingly demonstrate an important role for CD8⁺ T_{EM} cells in metabolic memory.

The findings of Garcia et al. raise several important questions. Although multiple studies, including the present

one, have demonstrated clonal expansion of specific T-cell populations in obesity (17,18), the identity of the antigens and the APC types responsible for this selection, particularly under WC conditions, remains incompletely understood. Several recent studies have proposed candidate antigens in obese adipose tissue, including lactate dehydrogenase A-derived peptides (19) and isolevuglandin (18). Regarding antigen presentation to CD8⁺ T cells, adipocytes have been shown to present antigen to CD8⁺ T cells (14). Direct *in vivo* evidence for dendritic cell- or macrophage-mediated CD8⁺ T-cell priming in adipose tissue is limited, but dendritic cells, classical cross-presenters, remain the most plausible APCs. Resolving which antigens are dominant and which APCs present them *in vivo* also will be crucial for mechanistic insight. In addition, while visceral depots such as epididymal white adipose tissue (WAT) have been the primary focus of most studies, subcutaneous depots such as inguinal WAT, generally considered less inflammatory, have also been reported to accumulate memory T cells following prolonged high-fat diet feeding (17). Determining how memory T-cell formation in different adipose depots affects systemic metabolic phenotypes will broaden our understanding of immunometabolic regulation. Moreover, validating these findings to human studies will be essential for translating the concept of targeting metabolic memory into viable therapeutic strategies.

Collectively, this study offers deeper insight into the mechanisms underlying worsened metabolic phenotypes during WC. These findings significantly advance our understanding of how metabolic memory of prior obesity can be stored and subsequently shape adipose tissue inflammation and systemic metabolic homeostasis. By elucidating these processes, this work lays a foundation for the development of

therapeutic strategies aimed at resetting metabolic memory and preventing dysfunctional metabolic outcomes during repeated cycles of WL and regain.

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